Differential Geometry: Key Definitions, Methods, and Examples

Generated Summary

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Contents

1 Curves in $\mathbb{R}^3$

1.1 Basic Definitions of curves

- **Length of a Curve:** The length of a parameterized curve c is $L(c) := \int_a^b \|\dot{c}(t)\|\,\mathrm{d}t$.
- **Regular Curve:** A curve $c(t)$ is regular if $\dot{c}(t) \neq 0$ for all $t \in I$.
- **Arc Length Parametrization**: A curve $c(s)$ is unit-speed if $\|\dot{c}(s)\| = 1$.

  For exerty regular curve, $\psi(t) = \int_{t_0}^t \|\dot{c}(t)\|\,\mathrm{d}t, \psi^{-1} \equiv \varphi, \tilde{c} = c \circ \varphi$ is unit-speed.
- **Frenet Frame:** If c is a unit-speed curve, $T(t) := \dot{c}(t), N(t) := \dfrac{\ddot{c}(t)}{\|\ddot{c}(t)\|}$ has unit length nad are perpendicular to each other.
- **Curvature (κ)**: Measures how fast the curve turns. For a unit-speed curve, $\kappa(t) = \|\ddot{c}(t)\|$.

  For a general curve, $\kappa(t) = \dfrac{\|\dot{c}(t) \times \ddot{c}(t)\|}{\|\dot{c}(t)\|^3} = \dfrac{\sqrt{\|\ddot{c}(t)\|^2 - \dot{v}(t)^2}}{v(t)^2} = \dfrac{\sqrt{v(t)^2\|\ddot{c}(t)\|^2 - (\dot{c}(t)\cdot\ddot{c}(t))^2}}{v(t)^3}$
- **Torsion (τ)**: Measures how much the curve twists out of the plane of curvature.

  For a general curve, $\tau(t) = \dfrac{(\dot{c}(t) \times \ddot{c}(t)) \cdot \dddot{c}(t)}{\|\dot{c}(t) \times \ddot{c}(t)\|^2}$.
- **Frenet-Serret Frame**: Let c be a unit-speed curve in $\mathbb{R}^3$, $T(t) = \dot{c}(t), N(t) = \dfrac{\ddot{c}(t)}{|\ddot{c}(t)|} = \dfrac{\ddot{c}(t)}{|\kappa(t)|}, B(t) = T(t) \times N(t)$ are the tangent field, the normal field and the binormal field.$\{T, N, B\}$ is called the Frenet-Serret frame.
- **Frenet-Serret Formulas**:

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix} \tag{1}$$

1.2 Extra Applications

- **Isoperimetric Inequality**: For a closed curve of length L enclosing area A, $L^2 \geq 4\pi A$ with equality iff the curve is a circle.
- **Winding Number**: Let c a loop and P_0 be a point not lying on the curve, the winding number $W(c, P_0)$ is the number of times c goes counter clockwise around P_0.

  If $c(t) = P_0 + r(t)(\cos\theta(t), \sin\theta(t))$ and $P_0 = 0$, then $W(c, P_0) = \dfrac{1}{2\pi}\displaystyle\int_a^b \frac{\dot{y}x - \dot{x}y}{x^2 + y^2}(t)\,dt$.

1.3 Calculation Examples

1.3.1 Example: Helix

Consider the helix $c(t) = (a\cos t, a\sin t, bt)$.

- **Velocity**: $\dot{c}(t) = (-a\sin t, a\cos t, b)$. Speed $v = \sqrt{a^2 + b^2}$.
- **Unit Tangent**: $T = \dfrac{1}{v}(-a\sin t, a\cos t, b)$.
- **Curvature**: $\kappa = \dfrac{a}{a^2 + b^2}$.
- **Torsion**: $\tau = \dfrac{b}{a^2 + b^2}$.

1.3.2 Example: Plane Curves

For a plane curve $c(t) = (x(t), y(t))$, the curvature is given by:

$$\kappa = \frac{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}{(\dot{x}^2 + \dot{y}^2)^{3/2}} \tag{2}$$

2 Surfaces in $\mathbb{R}^3$

2.1 Definition and Construction of Surfaces

- **Regular Map:**
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- **Coordinate Chart:** Given a subset $S \subseteq \mathbb{R}^3$, a coordinate chart on S around a point $m \in S$, consists of
  - an open subset $D \subseteq \mathbb{R}^2$
  - a regular,injective smooth map $\Phi : D \to S$, $\Phi(D) \subseteq S$
  - an open subset $\theta \in \mathbb{R}^3, m \in \theta$

  such that $\Phi(D) = S \cap \theta$ and $\Phi^{-1} : \theta \cap S \to D$ is continuous.
- **Surface:** A subset $S \subseteq \mathbb{R}^3$ is a surface if every point $m \in S$ has a coordinate chart centered at m. There are three common surfaces:
  - **Ruled Surface:** A ruled surface is a surface swept only by staight line $l : I \to \mathbb{R}^3$ moving along some curve $c : I \to \mathbb{R}^3 \Rightarrow \Phi(t, r) = c(t) + rl(t), r \in \mathbb{R}$
  - **Graphs:** Evert smooth function of $D \subseteq \mathbb{R}^2 \to \mathbb{R}^3$ determines a parameterized surface $S = \text{graph } f = \{(x, y, z) \in D \times \mathbb{R} : z = f(x, y)\}$
  - **Level Sets:** Let $g : \mathbb{R}^3 \to \mathbb{R}$ a smooth function.For every $c \in \mathbb{R}$, consider $S = g^{-1}(C) = \left\{(x, y, z) \in \mathbb{R}^3, g(x, y, z) = c\right\}$ is called a level set.

    If $\left(\frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z}\right) \neq 0$ on $g^{-1}(C)$, then S is a surface.
- **Tangent Plane**
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2.2 First Fundamental Form

- **Parametrization**: $\Phi(u, v) : U \to \mathbb{R}^3$. Tangent basis: $\{\Phi_u, \Phi_v\}$.
- **Coefficients**:

$$E = \langle \Phi_u, \Phi_u \rangle, \quad F = \langle \Phi_u, \Phi_v \rangle, \quad G = \langle \Phi_v, \Phi_v \rangle \tag{3}$$

- **Metric Matrix**: $I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$.
- **Arc Length on Surface**: $ds^2 = Edu^2 + 2Fdudv + Gdv^2$.
- **Area Element**: $dA = \sqrt{EG - F^2}\, du\, dv$.

2.3 Second Fundamental Form

- **Unit Normal**: $n = \dfrac{\Phi_u \times \Phi_v}{\|\Phi_u \times \Phi_v\|}$.
- **Coefficients**:

$$e = \langle \Phi_{uu}, n \rangle, \quad f = \langle \Phi_{uv}, n \rangle, \quad g = \langle \Phi_{vv}, n \rangle \tag{4}$$

- **Matrix**: $II = \begin{pmatrix} e & f \\ f & g \end{pmatrix}$.

2.4 Curvature

- **Shape Operator (Weingarten Map)**: $S = -dn$. Matrix representation:

$$S = I^{-1} II = \frac{1}{EG - F^2} \begin{pmatrix} G & -F \\ -F & E \end{pmatrix} \begin{pmatrix} e & f \\ f & g \end{pmatrix} \tag{5}$$

- **Gaussian Curvature (K)**: Determinant of Shape Operator.

$$K = \det(S) = \frac{eg - f^2}{EG - F^2} \tag{6}$$

- **Mean Curvature (H)**: Trace of Shape Operator.
$$H = \frac{1}{2}\mathrm{tr}(S) = \frac{eG - 2fF + gE}{2(EG - F^2)} \tag{7}$$
- **Principal Curvatures (k_1, k_2)**: Eigenvalues of S. $K = k_1 k_2$, $H = \dfrac{k_1 + k_2}{2}$.

2.5 Calculation Examples

2.5.1 Example: Cylinder

Parametrization: $\Phi(\theta, z) = (r\cos\theta, r\sin\theta, z)$.

- $E = r^2, F = 0, G = 1$.
- $e = -r, f = 0, g = 0$ (with inward normal).
- **Gaussian Curvature**: $K = \dfrac{(-r)(0) - 0}{r^2} = 0$.
- **Mean Curvature**: $H = -1/(2r)$.
- **Geodesics**: Helices, circles, and vertical lines.

2.5.2 Example: Sphere of Radius R

- $K = \dfrac{1}{R^2}$.
- $H = -\dfrac{1}{R}$ (or $1/R$ depending on normal orientation).

2.5.3 Example: Torus

Parametrization: $\Phi(u, v) = ((R + r\cos v)\cos u, (R + r\cos v)\sin u, r\sin v)$.

- **Gaussian Curvature**:
$$K = \frac{\cos v}{r(R + r\cos v)} \tag{8}$$
- Outer circle ($v = 0$): $K > 0$. Inner circle ($v = \pi$): $K < 0$. Top/Bottom ($v = \pm\pi/2$): $K = 0$.

2.5.4 Example: Surface of Revolution

Curve $c(t) = (r(t), 0, z(t))$ rotated around z-axis.

- **Gaussian Curvature**:
$$K = -\frac{r''(t)}{r(t)} \quad \text{(assuming unit speed profile)} \tag{9}$$

3 Moving Frames and Dual Frames (Cartan's Method)

3.1 General Theory for Orthonormal Frames

- **Frame and Coframe**: Let $\{e_1, \ldots, e_n\}$ be an orthonormal frame field ($\langle e_i, e_j\rangle = \delta_{ij}$). The **Dual Frame (Coframe)** is the set of 1-forms $\{\omega^1, \ldots, \omega^n\}$ such that:
$$\omega^i(e_j) = \delta^i_j \tag{10}$$
Any 1-form α can be written as $\alpha = \sum \alpha(e_i)\omega^i$.
- **Connection Forms**: The 1-forms ω_{ij} define the covariant derivative of the frame:
$$\nabla e_i = \sum_j \omega_{ji} e_j \quad \text{or equivalently} \quad de_i = \sum_j \omega_{ji} e_j \tag{11}$$
Condition: For orthonormal frames, the connection matrix is skew-symmetric:
$$\omega_{ij} = -\omega_{ji} \quad (\implies \omega_{ii} = 0) \tag{12}$$

3.2 Cartan's Structure Equations

These equations hold for any orthonormal frame in a Riemannian manifold (e.g., $\mathbb{R}^3$ or a surface).

1. **First Structure Equation** (Torsion-Free Condition): Relates the exterior derivative of the coframe to the connection forms.
$$d\omega^i = \sum_j \omega^j \wedge \omega_{ji} = -\sum_j \omega^j \wedge \omega_{ij} \tag{13}$$
Usage: Used to determine ω_{ij} from known ω^i.

2. **Second Structure Equation** (Curvature): Relates the curvature 2-forms Ω_{ij} to the connection forms.
$$\Omega_{ij} = d\omega_{ij} + \sum_k \omega_{ik} \wedge \omega_{kj} \tag{14}$$
For flat space ($\mathbb{R}^3$), $\Omega_{ij} = 0$, so $d\omega_{ij} = -\sum \omega_{ik} \wedge \omega_{kj}$.

3.3 Application to Surfaces in $\mathbb{R}^3$

Let $S \subset \mathbb{R}^3$ be a surface. Choose an **Adapted Frame** $\{e_1, e_2, e_3\}$ such that e_1, e_2 are tangent to S and e_3 is the unit normal (n).

- **Restriction to Surface**: On S, the normal displacement is zero, so:
$$\omega^3 = 0 \tag{15}$$
- **Connection Forms on Surface**: From $d\omega^3 = \omega^1 \wedge \omega_{13} + \omega^2 \wedge \omega_{23} = 0$, we deduce relations involving the Second Fundamental Form coefficients h_{ij}:
$$\omega_{13} = h_{11}\omega^1 + h_{12}\omega^2, \quad \omega_{23} = h_{21}\omega^1 + h_{22}\omega^2 \quad (h_{12} = h_{21}) \tag{16}$$
- **Gaussian Curvature (K)**: The intrinsic curvature of the surface is determined by the connection form ω_{12} (tangent rotation):
$$d\omega_{12} = -K\,\omega^1 \wedge \omega^2 \tag{17}$$
Alternatively, using the Gauss Equation: $d\omega_{12} = -\omega_{13} \wedge \omega_{32}$.
- **Mean Curvature (H)**:
$$H = \frac{1}{2}(h_{11} + h_{22}) \tag{18}$$
where $\omega_{13} \wedge \omega_{23} = K\omega^1 \wedge \omega^2$ (determinant) and $\omega_{13} \wedge \omega^2 + \omega^1 \wedge \omega_{23} = 2H\omega^1 \wedge \omega^2$ (trace).

3.4 Algorithm: Computing Curvature via Coframe

1. **Choose a Frame**: Find orthonormal e_1, e_2 tangent to S.
2. **Find Coframe**: Write $ds^2 = (\omega^1)^2 + (\omega^2)^2$. Identify ω^1, ω^2.
3. **Differentiate**: Compute $d\omega^1$ and $d\omega^2$.
4. **Solve for ω_{12}**: Use the First Structure Equation:
$$d\omega^1 = \omega^2 \wedge \omega_{21}, \quad d\omega^2 = \omega^1 \wedge \omega_{12}$$
(Recall $\omega_{21} = -\omega_{12}$).
5. **Compute K**: Calculate $d\omega_{12}$ and write it as $-K\omega^1 \wedge \omega^2$.

3.5 Example: Spherical Coordinates

On a sphere of radius R:

- **Coframe**: $\omega^1 = Rd\theta$, $\omega^2 = R\sin\theta d\phi$.
- **Differentials**: $d\omega^1 = 0$, $d\omega^2 = R\cos\theta d\theta \wedge d\phi = \dfrac{\cos\theta}{R\sin\theta}\omega^1 \wedge \omega^2$.
- **Connection Form**: Assume $\omega_{12} = A\omega^1 + B\omega^2$. $0 = d\omega^1 = \omega^2 \wedge (-\omega_{12}) \implies \omega_{12} \propto \omega^2$. $d\omega^2 = \omega^1 \wedge \omega_{12}$. Result: $\omega_{12} = -\cos\theta d\phi$.
- **Curvature**: $d\omega_{12} = \sin\theta d\theta \wedge d\phi = \dfrac{1}{R^2}\omega^1 \wedge \omega^2 \implies K = \dfrac{1}{R^2}$.

3.6 The Darboux Frame (Frame along a Curve)

For a unit-speed curve $c(s)$ on a surface S:

- **Definition**: An orthonormal frame $\{T, V, n\}$ adapted to both the curve and the surface.
  - $T(s) = \dot{c}(s)$ (Unit Tangent).
  - $n(s)$ is the surface unit normal restricted to the curve.
  - $V(s) = n(s) \times T(s)$ (Tangent Normal / Side Normal).
- **Darboux Equations** (Analogous to Frenet formulas):

$$\begin{pmatrix} T' \\ V' \\ n' \end{pmatrix} = \begin{pmatrix} 0 & \kappa_g & \kappa_n \\ -\kappa_g & 0 & \tau_g \\ -\kappa_n & -\tau_g & 0 \end{pmatrix} \begin{pmatrix} T \\ V \\ n \end{pmatrix} \tag{19}$$

- **Curvatures**:
  - **Geodesic Curvature** (κ_g): Bending in the tangent plane. $\kappa_g = \langle T', V\rangle = \omega_{12}(T)$.
  - **Normal Curvature** (κ_n): Bending normal to the surface. $\kappa_n = \langle T', n\rangle = II(T,T) = \omega_{13}(T)$.
  - **Geodesic Torsion** (τ_g): Twisting of the normal. $\tau_g = \langle n', V\rangle = \omega_{23}(T)$.
- **Relation to Space Curvature**: The curvature vector of the curve is $\vec{\kappa} = T' = \kappa_g V + \kappa_n n$. Squared curvature: $\kappa^2 = \kappa_g^2 + \kappa_n^2$.

4 Geodesics

4.1 Definitions

- **Geodesic**: A curve $\gamma(t)$ with zero geodesic curvature ($\kappa_g = 0$). Equivalently, the acceleration is normal to the surface.
- **Geodesic Equations**:

$$\ddot{u}^k + \sum_{i,j} \Gamma^k_{ij}\dot{u}^i\dot{u}^j = 0 \tag{20}$$

where Γ^k_{ij} are Christoffel symbols derived from the metric g_{ij}.

4.2 Examples

- **Plane**: Straight lines.
- **Sphere**: Great circles.
- **Cylinder**: Helices, circles, lines.

5 Gauss-Bonnet Theorem

5.1 Local Gauss-Bonnet

For a region R with boundary ∂R:

$$\iint_R K\,dA + \int_{\partial R} \kappa_g\,ds + \sum \theta_i = 2\pi \tag{21}$$

where θ_i are exterior angles at vertices.

5.2 Global Gauss-Bonnet

For a compact surface S without boundary:

$$\iint_S K\,dA = 2\pi\chi(S) \tag{22}$$

where $\chi(S) = 2 - 2g$ is the Euler characteristic (g is the genus/number of holes).

5.3 Example: Holonomy on Sphere

Parallel transport of a vector around a latitude circle C at colatitude θ_0.

- Angle of rotation (Holonomy): $\Delta\phi = 2\pi\cos\theta_0$.
- This equals the integral of curvature over the cap bounded by C.